Symmetric-Style Oblivious Lattice Evaluation

Or: OPRFs for the symmetric cryptographer

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- I study lattice-based OPRFs from heuristic assumptions to get *efficient* OPRFs.
- Recently built an OPRF more efficient than elliptic curves! [HKL+25]
- Needs more analysis- especially the underlying PRF
  joint work with Sonja Denk (Master student @ TUG)

Motivation for Oblivious Pseudorandom Functions

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Pseudorandom Function

A pseudorandom function (PRF) [GGM84, GGM86] is a deterministic and polynomial time function $F\colon \{0,1\}^k \times \{0,1\}^x \to \{0,1\}^y$ such that there is no probabilistic polynomial-time algorithm to distinguish any output y from a randomly chosen element from $\{0,1\}^y$.

Definitions

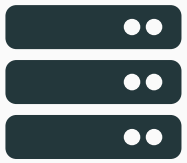
Client

input $c \in \mathcal{C}$, gets y

compute

$$y := F_k(c)$$

Server



key $k \in \mathcal{K}$

Oblivious Pseudorandom Function

An oblivious pseudorandom function (OPRF) [FIPR05] is a protocol between two parties. One party holds the secret key k and the other holds their secret input x. The OPRF privately realizes the joint computation outputting $F_k(x)$ for a PRF F to the party holding x, and nothing to the party holding k.

Additional Properties

Partial Obliviousness In a POPRF, the client submits a tag t in addition to the input x and gets the evaluation $y := F_k(t, x)$. t contains additional information, e.g. a validity period.

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Public Verifiability means we built a blind signature.

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    - vADDG [ADDG24]: proposed 80 bits of security; attack estimates 10
    - Replication and Polynomial encoding schemes [CKP+24]: $\mathbb{P}_{\text{successful forgery}} = 1$, linear time complexity

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  - [SW25] bring the attack of [AR25] under the birthday bound and make it appicable to more parameters. This breaks $\frac{3}{4}$ of the constructions in *Improved alternating-moduli PRFs* [APRR24]

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- Example: Learning with Rounding (LWR) PRFs
  - [BPR12]: modulus has to be exponential in input length
  - Spring PRF [BBL+15]: large modulus seems to be a proof artifact

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Just use AES?
Algebraic structure for Oblivious Evaluation

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Example (Naor-Reingold with seven input bits)

For example, for 0110111 we would compute the group action $g_0 * g_2 * g_3 * g_5 * g_6 * g_7$.

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Multiplication is addition of exponents!

$$g^a g^b = g^{a+b} \text{ for some generator } g$$

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  - Spoiler the end result will look a lot like a symmetric cipher!

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- Compose out of $\mod 2$ constructions: $q = 16 = 2 * 2 * 2 * 2$

Almost a symmetric cipher: Spring mod 16

Symmetric structure

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- Optional round function through CTR mode for longer outputs

Efficient Subset-Products

Let's zoom into the computation $\mod 2$

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- abstract idea: $g^a g^b \equiv g^{a+b}$

Efficient Subset-Sum in R_2^* via the exponent representation

- We need a generator $g = 3 \mod 257$
- Break R_2^* into cyclic components: $C_2^{n/4} \times C_4^{n/8} \times C_8^{n/16} \times \ldots \times C_n$

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  - parallelizable!

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  - Multiplying by $1 + (1+x)^j$

$$A(x) \oplus (A(x) \ll j)$$

  - multiply out components using shift-and-xor

Substitution

$$\mathcal{F}_{\mathbf{K}}(c_1, \cdots, c_m) = S\left(\text{iNTT}\left(\text{lookup}\left(\mathbf{k}_0 + \sum_{i=1}^{m} c_i \mathbf{k}_i\right)\right)\right)$$

- given a, compute $g^a \mod q$
- alternative mod 2: Exponent to radix representation
  - Multiplying $A(X)$ by $(1+x)^j$ means a leftshift by j.
  - Multiplying by $1 + (1+x)^j$

$$A(x) \oplus (A(x) \ll j)$$

  - multiply out components using shift-and-xor
    Final form: $(x+1)^j$

Permutation (Radix to power form)

$$\mathcal{F}_{\mathbf{K}}(c_1, \cdots, c_m) = S\left(\text{iNTT}\left(\text{lookup}\left(\mathbf{k}_0 + \sum_{i=1}^{m} c_i \mathbf{k}_i\right)\right)\right)$$

$$A(x) = c_0 * 1 + c_1 * (1 + x) + c_2 * (1 + x)^2 + ... + c_{127} * (1 + x)^{127}$$

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- Cooley-Tukey Radix 2 evluation:
  - instead of g we have the bases $\{1, (1 + x), (1 + x)^2, \ldots, (1 + x)^{127}\}$
  - $c_i \in \{0, 1\}$

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- Cooley-Tukey Radix 2 evluation:
  - instead of g we have the bases $\{1, (1+x), (1+x)^2, \ldots, (1+x)^{127}\}$
  - $c_i \in \{0, 1\}$
  - Same polynomial, different representation:

$$A(x)(1+x)^j = \sum_{i}^{127} c_i (1+x)^{i+j} \tag{1}$$

  - *Multiplying $A(X)$ by $(1+x)^j$ means a leftshift by j*
  - recursive operation: if
    exploit that $(1+x)^j \equiv 1 + x^j$
  - can also be done with shifts and masks

Truncation

$$\mathcal{F}_{\mathbf{K}}(c_1, \cdots, c_m) = S\left(\text{iNTT}\left(\text{lookup}\left(\mathbf{k}_0 + \sum_{i=1}^{m} c_i \mathbf{k}_i\right)\right)\right)$$

- Rounding function

$$\lfloor a \rceil_p = \begin{cases} 1 & \frac{q}{4} \leq a \geq \frac{3q}{4} \\ 0 & \text{otherwise} \end{cases}$$

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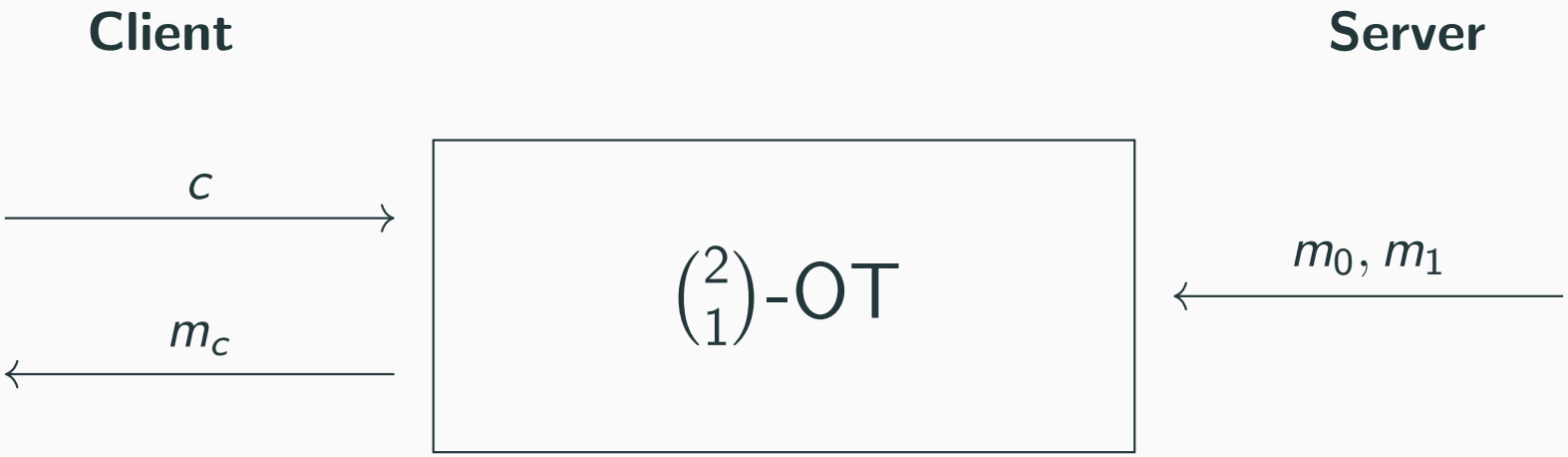
- alternatives possible, e.g. MSB rounding
- combine composition of results
- e.g. mod 16: four computations mod 2
- design is less clear

Just use AES?
Algebraic structure for Oblivious Evaluation

A little help from MPC friends



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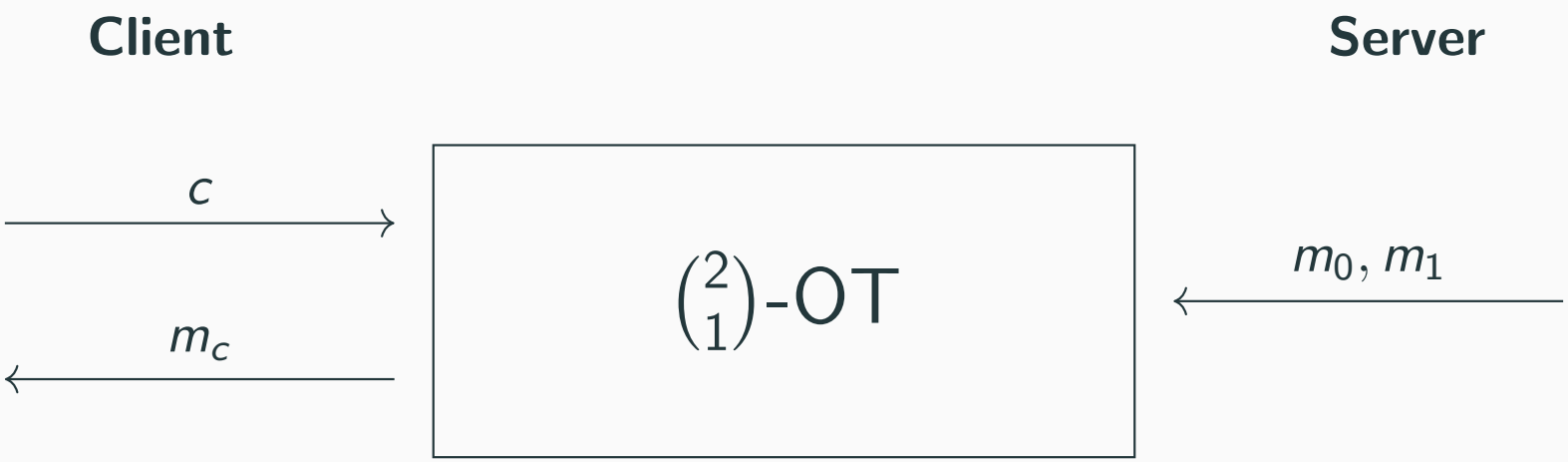
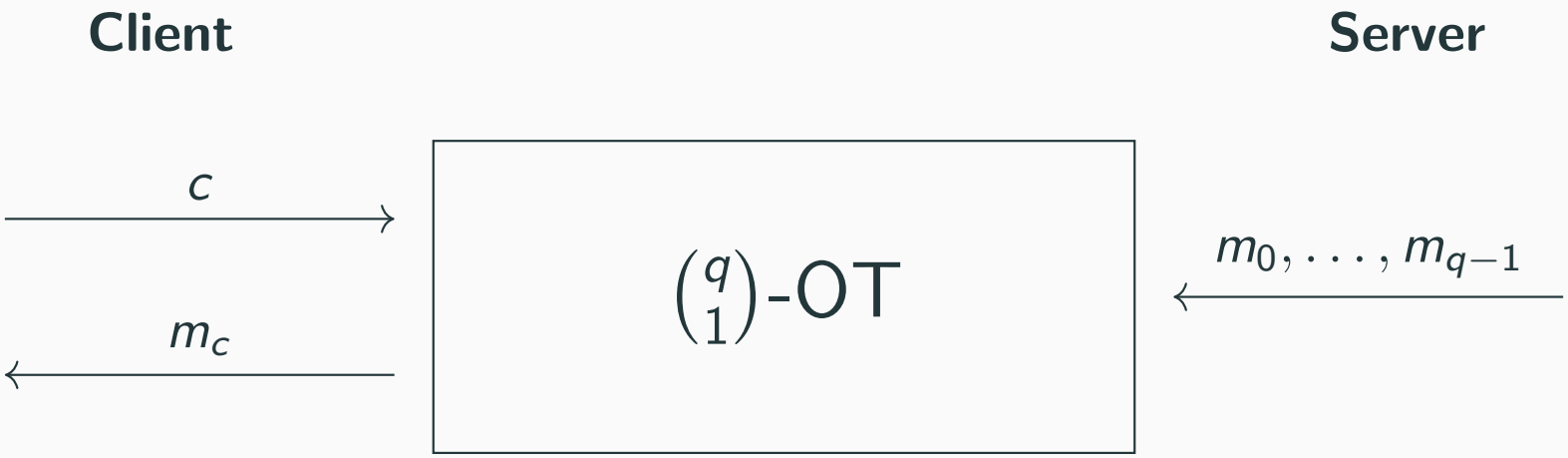


Extend with symmetric operations!

A little help from MPC friends (cont.)



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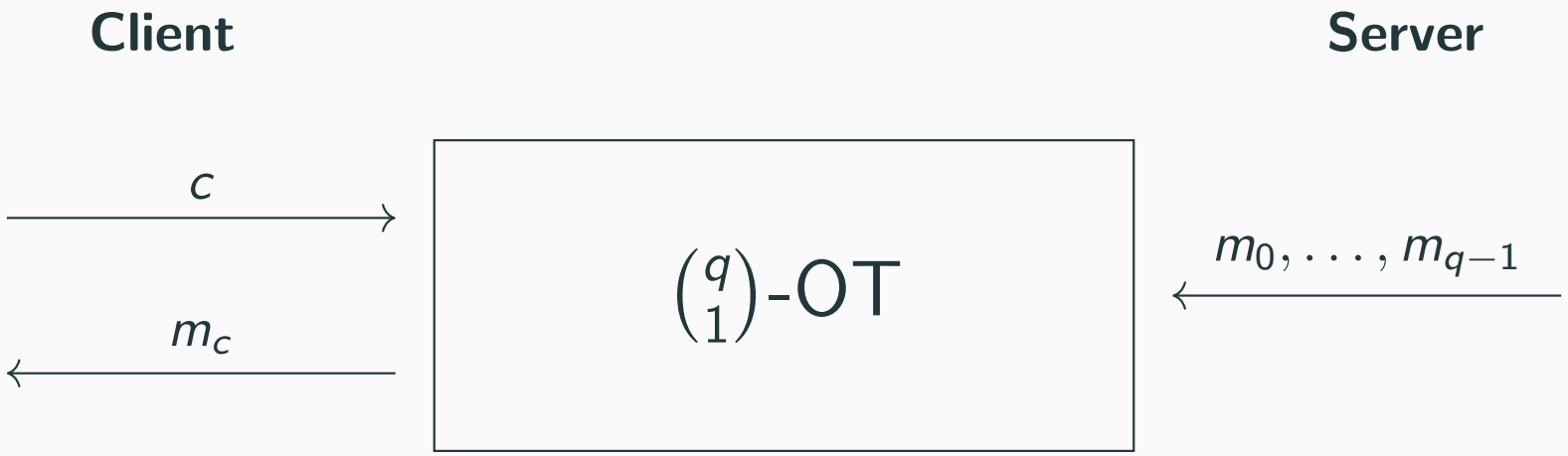
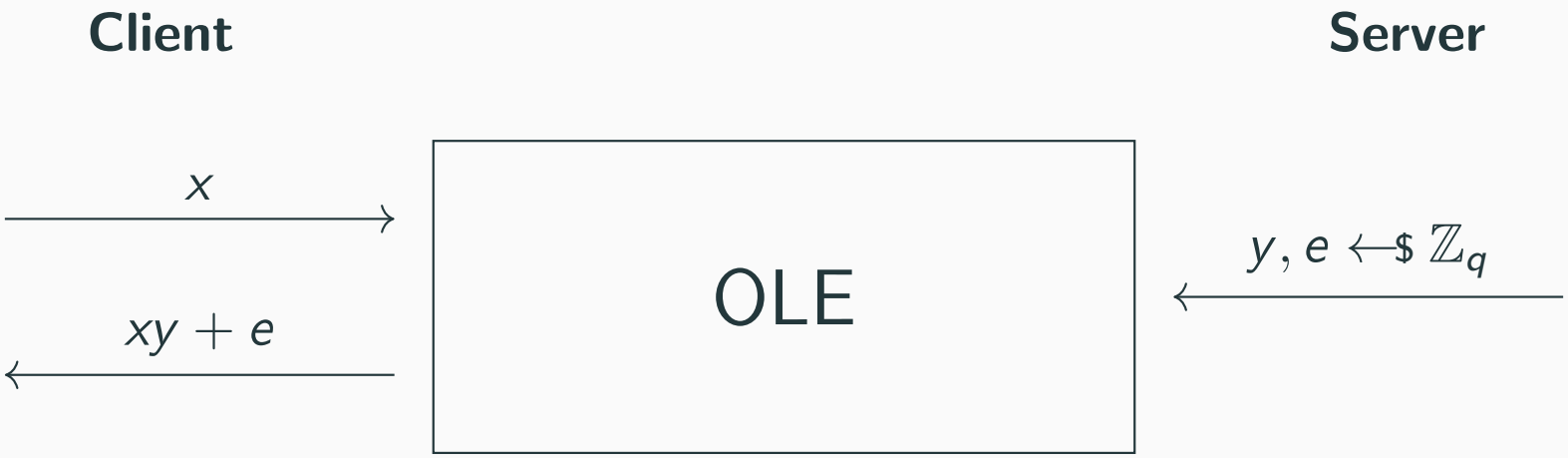
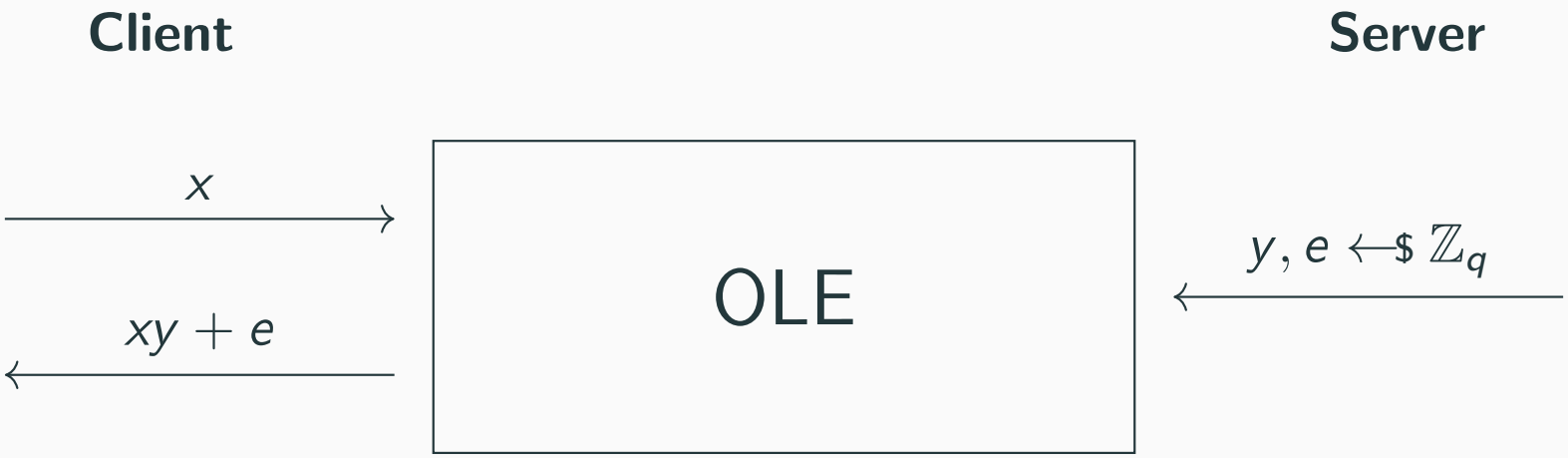


Generic construction from $\log q$ $\binom{2}{1}$-OTs !

A little help from MPC friends (cont.)



A little help from MPC friends (cont.)



Generic construction from $\log q$ $\binom{2}{1}$-OTs !

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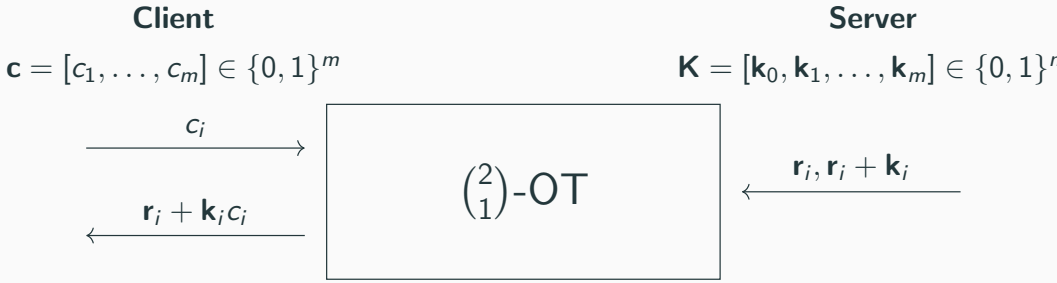
An OPRF from Spring



$$S\left(\text{iNTT}\left(\text{lookup}\left(\mathbf{k}_0 + \sum_{i=1}^{m} c_i \mathbf{k}_i\right)\right)\right)$$

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Client		Server
$\mathbf{c} = [c_1, \ldots, c_m] \in \{0,1\}^m$		$\mathbf{K} = [\mathbf{k}_0, \mathbf{k}_1, \ldots, \mathbf{k}_m] \in \{0,1\}^r$
$\xrightarrow{c_i}$	$\binom{2}{1}$-OT	$\xleftarrow{\mathbf{r}_i, \mathbf{r}_i + \mathbf{k}_i}$
$\xleftarrow{\mathbf{r}_i + \mathbf{k}_i c_i}$		
$\sum_{i=1}^{\kappa} \mathbf{r_i} + c_i \mathbf{k_i}$		$\mathbf{k}_0 - (\sum_{i=1}^{\kappa} \mathbf{r}_i)$

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$\mathbf{b} \leftarrow \prod_{i=1}^{\kappa} \mathbf{r_i} \mathbf{k_i}^{c_i}$		$\mathbf{a} \leftarrow \mathbf{k}_0 (\prod_{i=1}^{\kappa} \mathbf{r}_i)^{-1}$

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$\mathbf{y} \leftarrow \sum_{i=0}^{n-1} (a_i \cdot b_i + e_i) X^i$	iNTT	$\mathbf{e} \leftarrow \sum_{i=0}^{n-1} e_i X^i$
$\xrightarrow{y_i}$ $\xleftarrow{\lfloor y_i - e_i \rceil_p}$	$\binom{q}{1}$-OT	$\xleftarrow[\forall q_i \in \mathbb{Z}_q]{\lfloor (q_i - e_i) \rceil_p}$

An OPRF from Spring

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$\mathbf{y} \leftarrow \sum_{i=0}^{n-1} (a_i \cdot b_i + e_i) X^i$	iNTT	$\mathbf{e} \leftarrow \sum_{i=0}^{n-1} e_i X^i$ $\mathbf{d} \leftarrow\$ \{0,1\}^n$
$\xrightarrow{y_i}$ $\xleftarrow{\lfloor y_i - e_i \rceil_p \oplus d_i}$	$\binom{q}{1}$-OT	$\xleftarrow{\lfloor (q_i - e_i) \rceil_p \oplus d_i}$ $\forall q_i \in \mathbb{Z}_q$
	$\xleftarrow{\text{BCH}(\mathbf{d})}$	

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- Quesions? Ideas? Let's discuss!

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